

***WHY SHOULD WE CARE
ABOUT KERNELNEWBIES!***

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(@kernel_girl)**



Who Am I?

ORACLE®

Rails Girls
Summer of Code



OUTREACHY

Who is Kernelnewbie?

Type one:

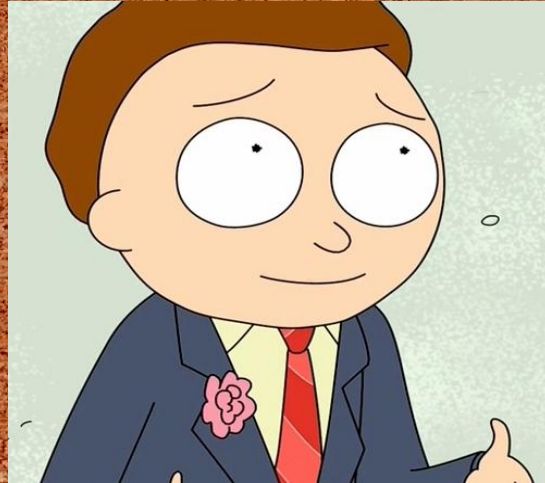
- Open source newbie
- May or may not have formal computer science education
- Curious, enthusiastic, confused



Who is Kernelnewbie?

Type two:

- Belongs to open source world, new to Linux kernel
- Wants to contribute to Linux Kernel out of interest or job
- Curious, Excited, confused



Flow of Morty's Journey in a Linux kernel world

1.

**Setting up an
environment for
the first
contribution**

2.

**First successful
contribution**

3.

**Find tasks/
projects
for more valued
contribution**

4.

**Quality
contribution
in the Linux
kernel / full-
time job**



*Why should you
care about
Morty's journey?*

- Fast kernel development process ->
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- Need of more subsystem maintainers/reviewers to share the load
- Need of more kernel developers
- Kernel community works on trust

Obstacles in Morty's journey

1: Setting up an environment

- First patch tutorial helps with setting up an environment but doesn't really talk about how kernel development process works
- It suggests checkpatch.pl fixes but not all subsystem maintainers are fine with accepting these fixes
- Different subsystems follows different workflow

Obstacles in Morty's journey

2: First successful contribution

- No specific time defined on when you will get reply or patch will get merged
- Emails == Chances of patches being lost
- Subsystem maintainers picks up the patches based on their work flow - no documentation on the same

Obstacles in Morty's journey

2: First successful contribution

- Patch tags - some prefers RFC and some don't
- Picky but not patient enough
- Rude behavior -> discourages Morty to continue their journey

Obstacles in Morty's journey

3: Find tasks for quality contribution

- Not using version control development platform == no open issues
- TODO files in subsystems are not often updated
- Hard to find TODO tasks listed as comments in files

Obstacles in Morty's journey

3: Find tasks for quality contribution

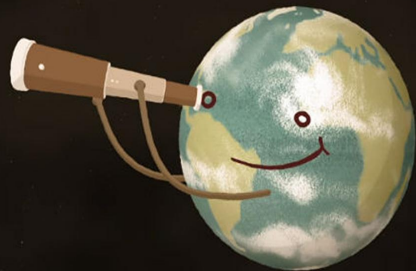
- Maintainers/developers often have few ideas but do not have time to implement them because of the busy schedule
- Newbies often look for more ideas but don't know where to find them

Obstacles in Morty's journey

4: Quality contribution in the kernel

- Many times new tasks often requires discussions with the people before sending patches
- Mailing lists/IRCs works for some subsystems but having an information on whom to reach out can help when starting

Kernel Planet



Possible Solution

Possible Solution



***Idea: A home to
address the
concerns***

- **Website/Wiki**
- **Involve kernel
developers/maintainers
who are interested in
building a bridge between
kernel newbies and kernel
developers / maintainers**

Things to be implemented

- List of subsystems who are fine with receiving checkpatch.pl fixes and guiding kernelnewbies in their journey of first perfect kernel patch

Things to be implemented

- **List of documentation available for each subsystem/kernel areas in a more organized manner**

Things to be implemented

- Subsystem wise TODO list of tasks (Similar to github's help needed/beginner friendly tags)

Things to be implemented

- Subsystem wise assigned list of reviewers or developers who wants to help in reviewing patches

Things to be implemented

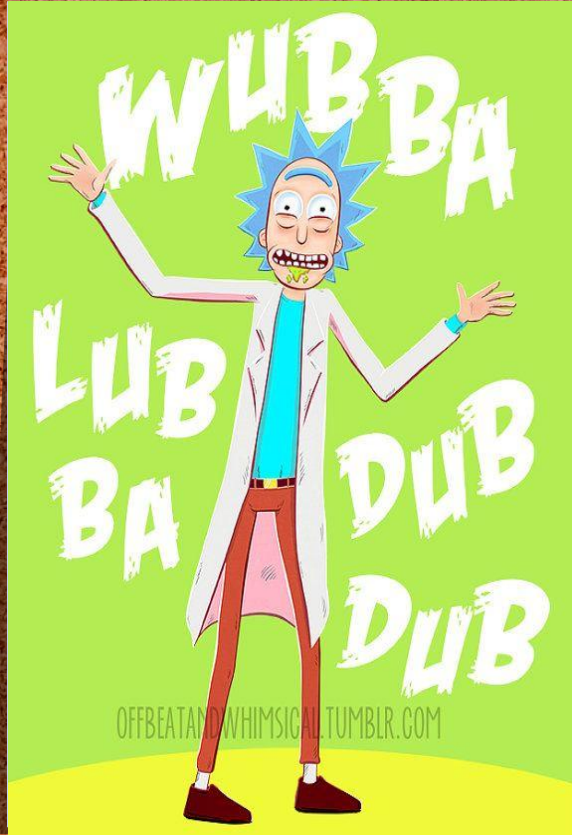
- **List of more ambitious projects [Just a bunch of possible ideas from developers/maintainers]**

*[RFC] Solutions
I'm not sure
about*

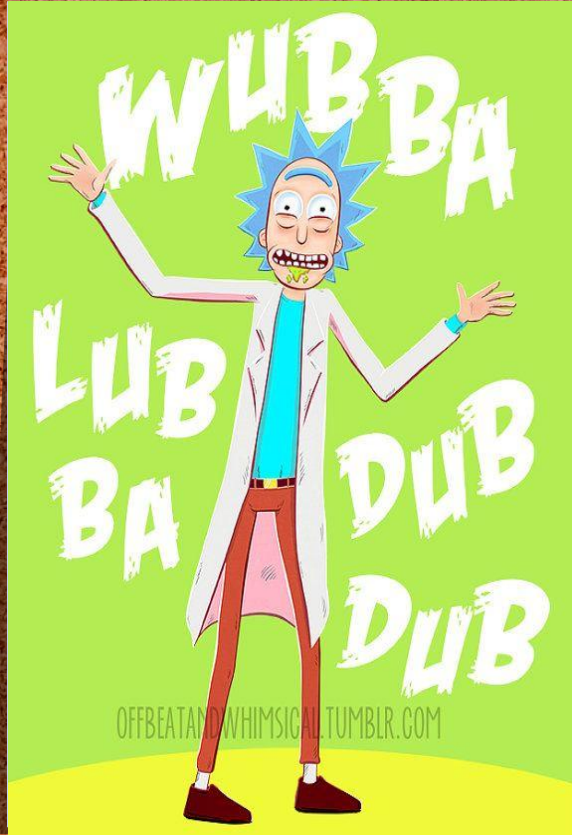
- **Problem:** Behaviour
- **Solution:** Code of conduct, basically something that says you'll not shout if the patch takes multiple revisions and bit of a time

*[RFC] Solutions
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about*

- **Problem:** No docs on subsystem wise development process
- **Solution:** Document the subsystem wise process of sending patch to patch being merged [approx time?]



More ideas?



More ideas?

[<https://github.com/nerdyvaishali/kernelbridge>]

Resources

- **Google.com**
- **Adult Swim - Rick and Morty**